

Shock-Chemistry Interaction in Supersonic Hydrocarbon Reacting Flows

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CENTRAL FINDING

A significant drift velocity develops between n-dodecane and surrounding nitrogen within the shock front, causing local fuel-lean/fuel-rich separation. Mach 10 shock compression and rapid thermal non-equilibrium drive progressive n-dodecane dissociation into light hydrocarbons dominated by ethylene.

01 Background + Method

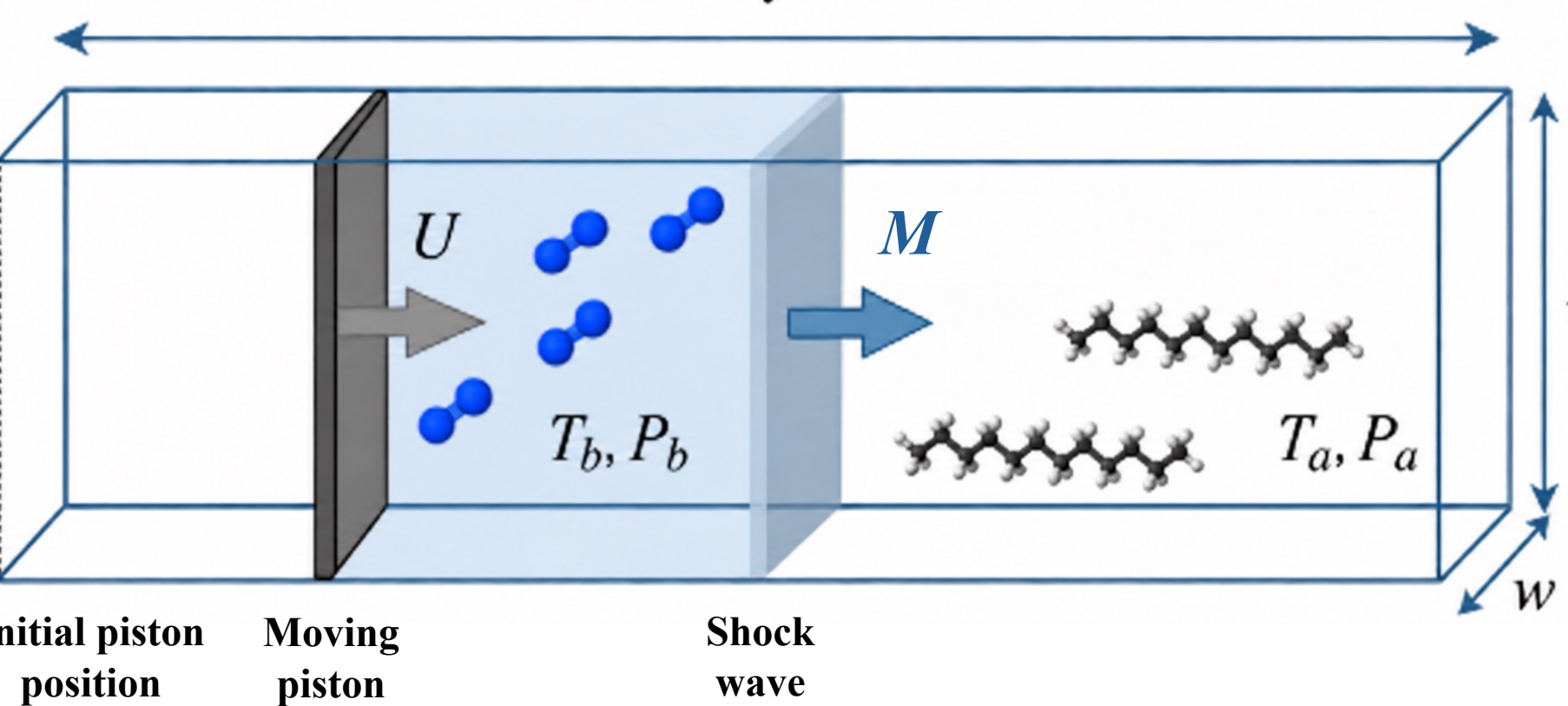
Why investigate shock chemistry using reactive forcefield?

High-Speed Combustion Challenges: Modeling jet fuel pyrolysis in extreme regimes (e.g., scramjet combustors at Mach 6–12) is critical for reactive flow CFD mechanisms, but experimental probing is difficult and quantum mechanics methods are too computationally expensive for dynamic, multi-molecule systems.

Limitations of Non-Reactive MD & Continuum Models: Continuum simulations assume local thermal equilibrium, while classical non-reactive forcefields (L-OPLS) feature rigid molecular connectivity, preventing them from capturing dynamic bond dissociation, chemical reactivity, or product formation.

Role of Reactive Molecular Dynamics (ReaxFF): ReaxFF incorporates variable bond orders along with long-range van der Waals and Coulombic interactions, enabling atomistic tracking of shock-driven initiation reactions, thermal non-equilibrium relaxation, and fuel decomposition.

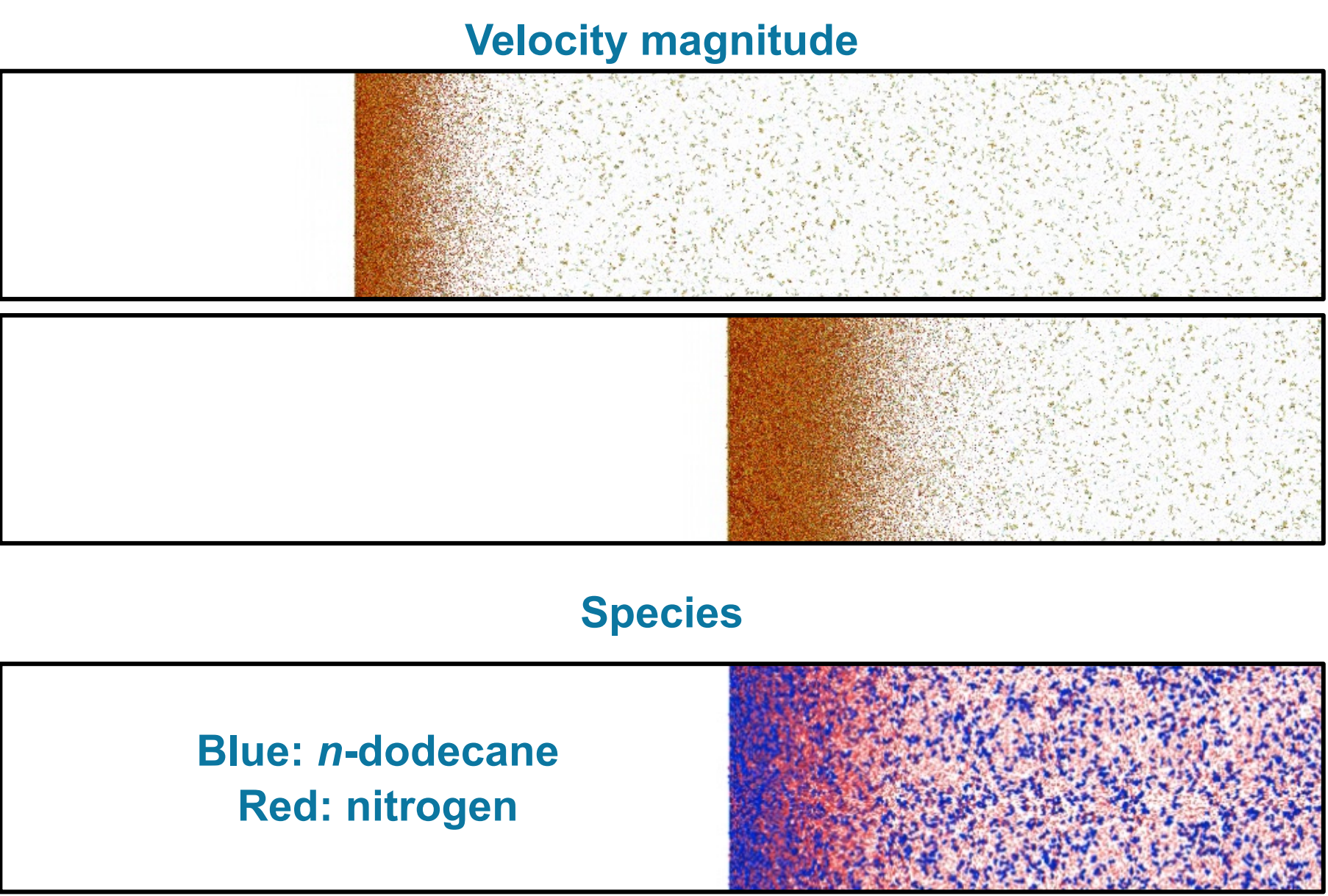
Computational setup



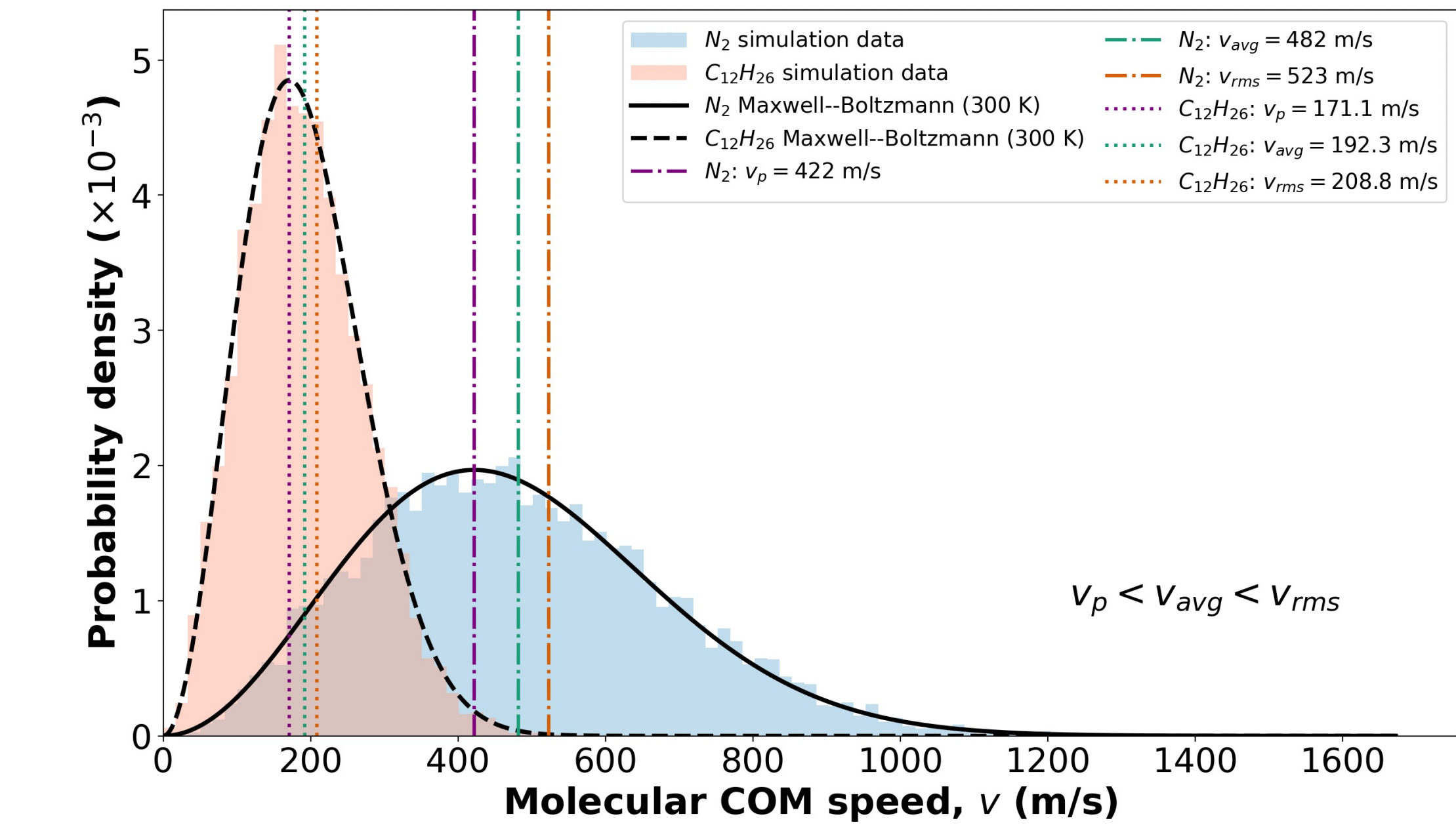
Computational workflow

- Uniform $N_2 / C_{12}H_{26}$ mixture, $X_{N_2} : X_{C_{12}H_{26}} = 88 : 1$
- Equilibrate the pre-shock state, $T_a = 300$ K, $P_a = 5$ atm under NVT then NVE
- Modeled using 2 forcefields:
L-OPLS (non-reactive), and ReaxFF (reactive)
- Induced Mach 5 and 10 shock waves using a moving piston

Representative MD snapshots



02 Maxwell-Boltzmann speed distribution at pre-shock state

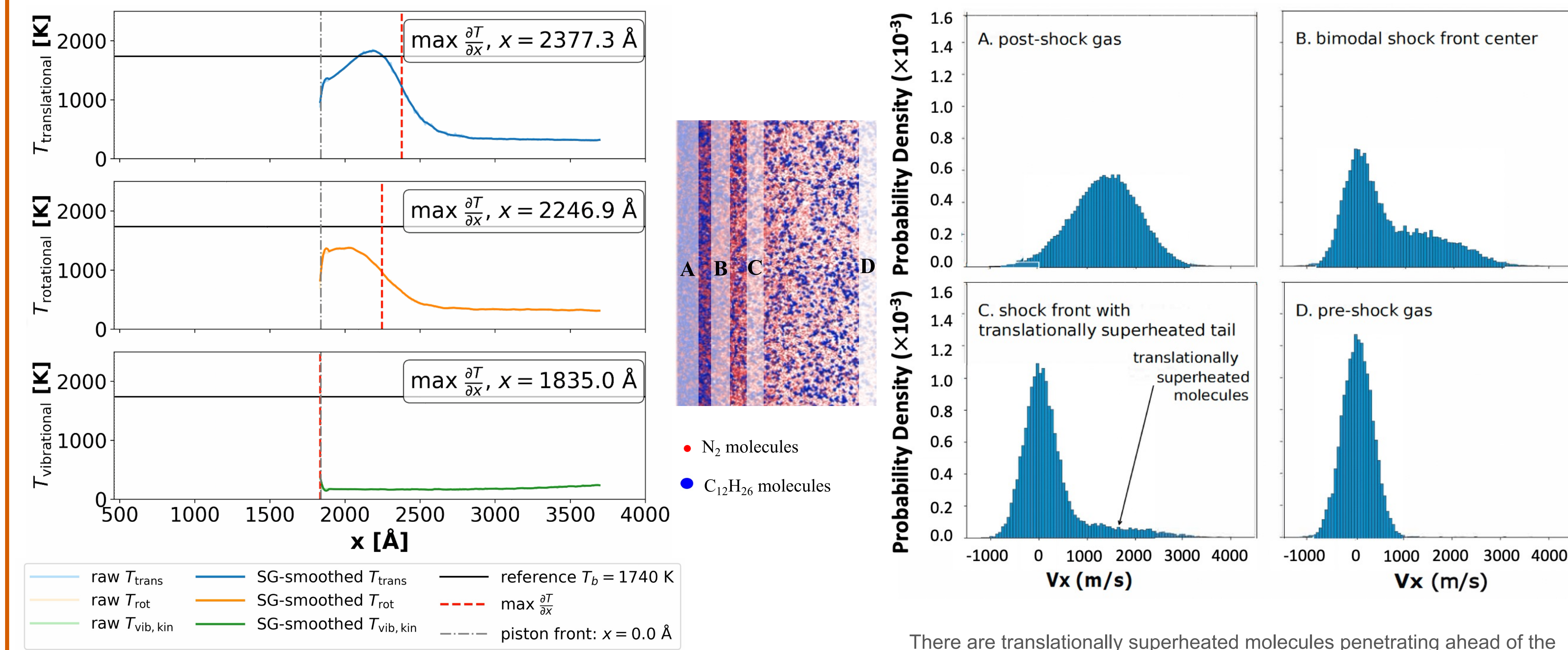


Maxwell-Boltzmann velocity distribution:

$$f(v) = 4\pi v^2 \left(\frac{m}{2k_B T} \right)^{3/2} e^{-\frac{mv^2}{2k_B T}}$$

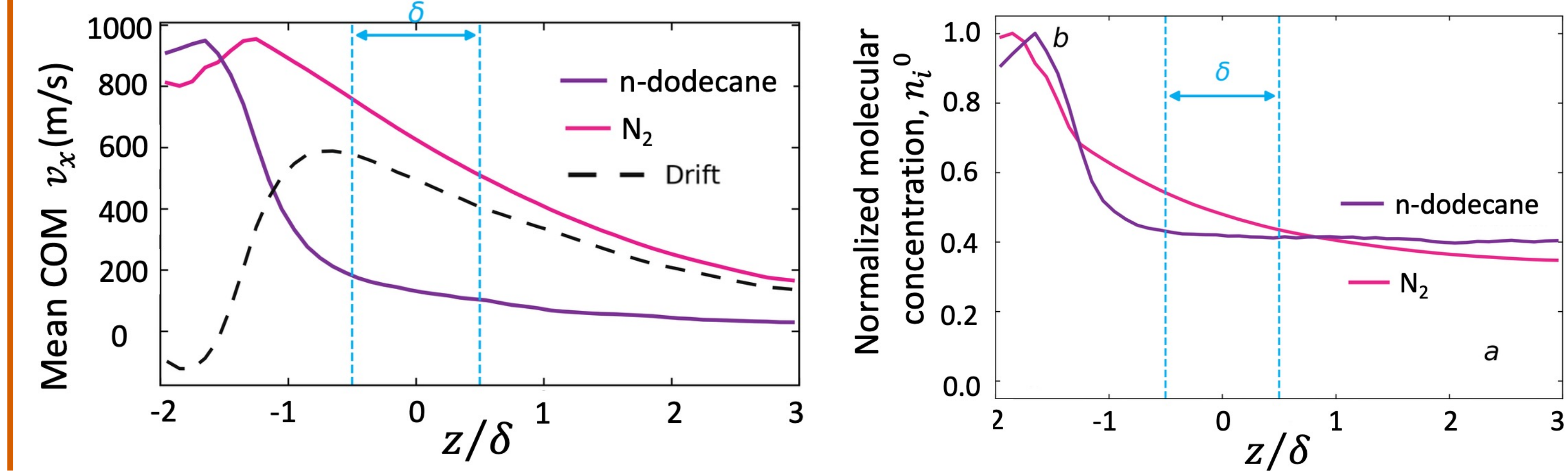
Both initial N_2 and $C_{12}H_{26}$ velocity distribution matches the theoretical Maxwell-Boltzmann distribution at 300 K, establishing a fully equilibrated thermodynamic baseline prior to shock stage.

03 Thermal Non-Equilibrium Profiles & Velocity Distributions (Mach 5)



There are translationally superheated molecules penetrating ahead of the nominal shock front. It promotes super-collisions with upstream molecules while broadening the effective shock thickness.

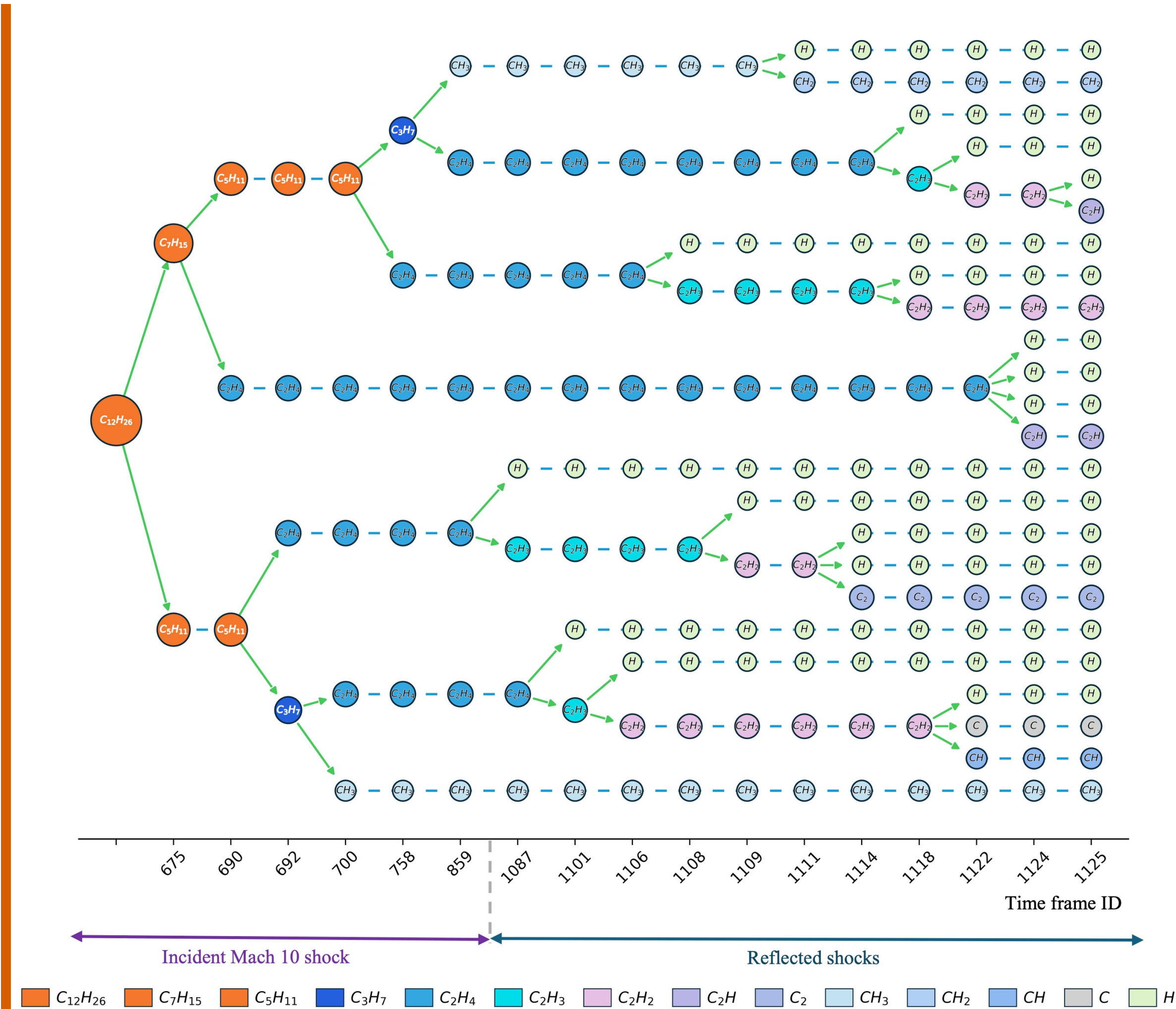
Strong thermal non-equilibrium develops across the normal shock. Directed bulk kinetic energy is rapidly converted into translational energy, producing a longitudinal temperature overshoot.



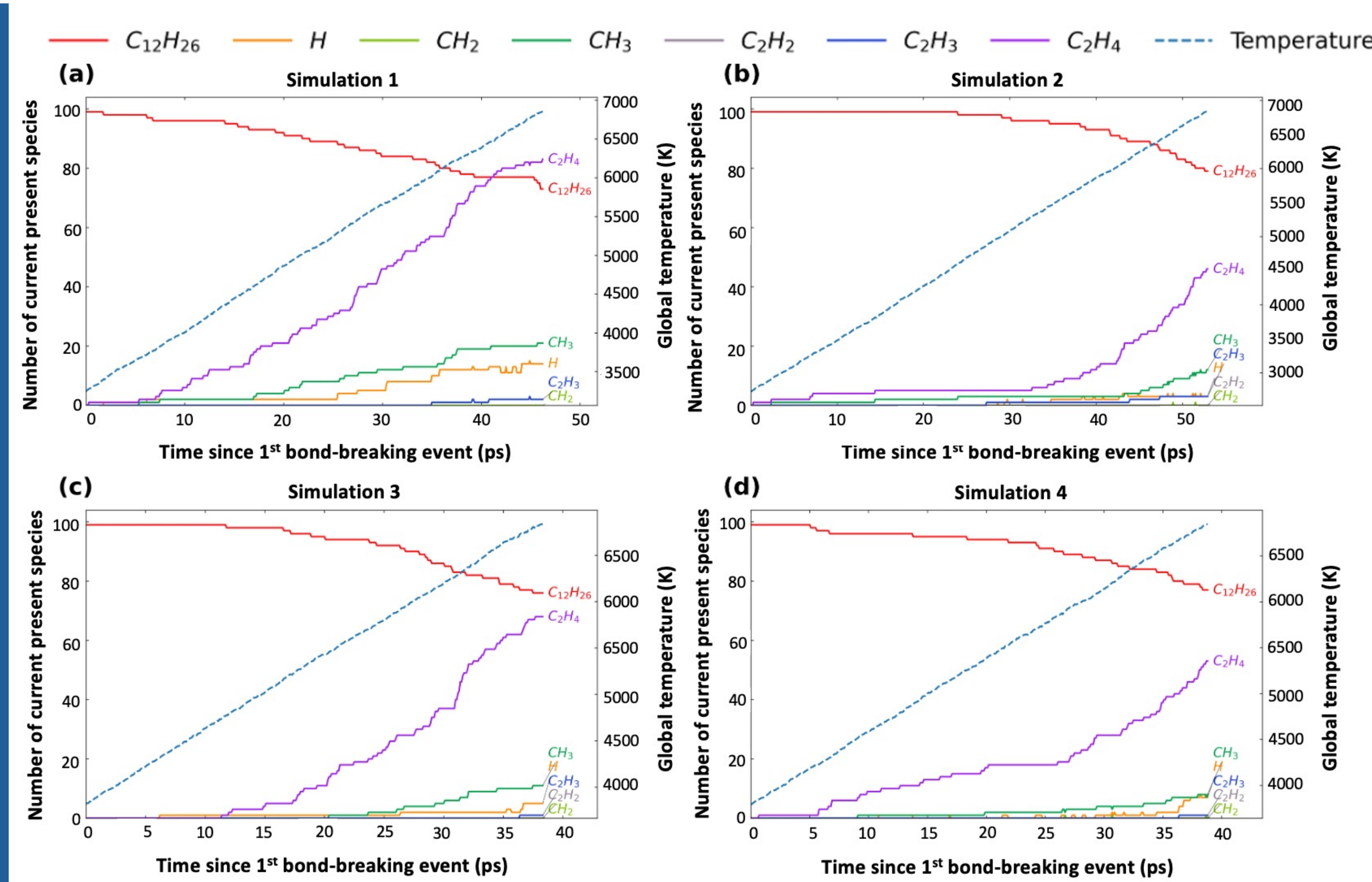
A significant drift velocity develops between the fuel and surrounding nitrogen, causing species separation within the shock front, with nitrogen accumulating ahead of n-dodecane and producing locally fuel-lean and fuel-rich regions in an initially premixed mixture.

04 Bond breaking and formation events (Mach 5 and 10)

Example fragmentation tree



Species evolution



Four independent shock MD simulations were performed. At Mach 10, bond-breaking and reformation would eventually convert all initially intact n-dodecane molecules into fragments, with C_2H_4 emerging as the dominant hydrocarbon product. C_2H_4 is observed primarily behind the shock front, consistent with post-shock fuel decomposition.

Acknowledgments

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